There’s a famous equation in astrobiology called the Drake Equation. In theory, it’s a way to calculate the number of civilizations in our galaxy with whom we might be able to communicate. All you have to do is input the number of stars, the fraction of those with planets, the fraction of those with life, the fraction with *intelligent* life, and so on, and in the end you get the number of messages you should expect on your interstellar voicemail. Of course, many of these input numbers are, at least with current data, completely impossible to determine, which means that the final answer isn’t meaningful. The thing that’s useful about the Drake Equation is that it makes us think about our *assumptions* about extraterrestrial life, and to figure out what we do and don’t know about this whole question.

Talking with Hiranya Peiris, it occurs to me that contemplating our ultimate cosmic destruction might be much the same. I suggest to her that perhaps we’re doing a calculation where the final number doesn’t matter, but the calculation does. “The number doesn’t matter,” she agrees, “but the exercise of thinking through the different options on the table, I think, is good.” And the implications of this thought experiment might ultimately pay off. “It could lead to some cool way to test between the hypotheses that doesn’t wait for seven billion years.”

How long *do* we have to wait for a breakthrough? We don’t (and can’t) know. We’re exploring off the edge of the map now. Clifford Johnson is very optimistic that we’re heading toward a better, deeper understanding of physics, but he acknowledges the caveat. “It might be that we go for a couple hundred years gathering all of this data before we see the signal and then we go back and realize that, oh, it was there staring us in the face all along. That’s an annoying possibility. But for questions as big as the ones we’re trying to answer, I feel that that’s okay. Why need it be of the length scale of a human lifetime?”

In the meantime, we’ll continue on, making new paths through the woods to see what we might find hiding there. Someday, deep in the unknown wilderness of the distant future, the Sun will expand, the Earth will die, and the cosmos itself will come to an end. In the meantime, we have the entire universe to explore, pushing our creativity to its limits to find new ways of knowing our cosmic home. We can learn and create extraordinary things, and we can share them with each other. And as long as we are thinking creatures, we will never stop asking: “What comes next?”